

PROJECT DOCUMENTATION PHASE

1. Phase Status Sheets (PSS) Compilation

The Phase Status Sheets (PSS) consolidate all technical descriptions, architectural designs, and system workflows into the official project documentation template:

- **Technical Descriptions:** Documents system goals, modular MVC architecture, JWT authentication flow, Mongoose ORM database integration, and Google Gemini AI service logic.
- **Entity-Relationship (ER) Diagram:** Details the One-to-Many (User → Workouts) database model, primary keys (`_id`), foreign key references (`user`), attribute data types, and validation constraints.
- **Data Flow Diagrams (DFD) & User Flow:** Charts the flow of incoming HTTP requests from API consumers through Express routes, JWT auth middleware, controllers, database models, and external Google Gemini API services.
- **Document Integration:** Assembles individual phase outputs — from initial Brainstorming & Ideation through Testing — into the final PSS report template.

2. Source Code Repository Setup

A. Clean GitHub Repository Structure

The GitHub repository organizes backend source code inside a dedicated `/server` directory to maintain modular separation (see the Backend Structure diagram in the Project Development Phase).

B. Strict File Exclusion (`.gitignore`)

To avoid committing bulky build dependencies and leaking confidential security keys, the `.gitignore` file strictly excludes `node_modules/` and local `.env` configuration files:

```
# Dependency directory - MUST BE EXCLUDED
node_modules/

# Environment and secret keys
.env
.env.local

# Logs and runtime data
npm-debug.log*
yarn-debug.log*

# OS metadata
.DS_Store
Thumbs.db
```

C. Project README.md Documentation

FitTrack AI - Backend RESTful API

FitTrack AI is a secure, AI-augmented RESTful backend built using Node.js, Express.js, MongoDB, and Google Gemini AI.

Prerequisites

- Node.js: v16.x or higher
- npm: v8.x or higher
- MongoDB: Local MongoDB instance or MongoDB Atlas URI
- Google Gemini API Key: Obtained from Google AI Studio

Installation & Setup

1. Clone the Repository:

```
git clone https://github.com/gnna-m/fittrack.git  
cd fittrack/server
```

2. Install Dependencies:

```
npm install
```

3. Configure Environment Variables:

Create a .env file inside the /server folder (see Environment Setup & Configuration Plan, Project Planning Phase).

4. Run the Server:

```
npm run dev
```